$\mathcal{K}$	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1}{}_{\alpha\beta\chi}$	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{K}_{1^-}^{\#1}{}_{\alpha}$	$\mathcal{K}_{1^-}^{\#2}{}_{\alpha}$	$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{2^-}^{\#1}{}_{\alpha\beta\chi}$
$\mathcal{K}_{0^+}^{\#1\,\dagger}$	$\frac{1}{2}(Y_1 - Y_2 k^2)$	0						
$\mathcal{K}_{0^-}^{\#1\,\dagger\alpha\beta\chi}$	0	$Y_3 + Y_4 k^2$						
$\mathcal{K}_{1^+}^{\#1\,\dagger\alpha\beta}$			$\frac{1}{2}(Y_5 + Y_6 k^2)$	$\frac{-Y_7 k^2 - 2\,\overset{(2)}{\kappa}_2}{2\sqrt{2}}$	0	0		
$\mathcal{K}_{1^+}^{\#2\,\dagger\alpha\beta}$			$\frac{-Y_7 k^2 - 2\,\overset{(2)}{\kappa}_2}{2\sqrt{2}}$	$Y_8 k^2 + \overset{(2)}{\kappa}_1$	0	0		
$\mathcal{K}_{1^-}^{\#1\,\dagger\alpha}$			0	0	$\frac{1}{2}(Y_9 + Y_{10} k^2)$	$\frac{Y_{11} k^2 + 2\,\overset{(2)}{\kappa}_3}{2\sqrt{2}}$		
$\mathcal{K}_{1^-}^{\#2\,\dagger\alpha}$			0	0	$\frac{Y_{11} k^2 + 2\,\overset{(2)}{\kappa}_3}{2\sqrt{2}}$	$\frac{1}{2}(Y_{12} + Y_{13} k^2)$		
$\mathcal{K}_{2^+}^{\#1\,\dagger\alpha\beta}$							$\frac{1}{2}(Y_{14} + Y_{15} k^2)$	0
$\mathcal{K}_{2^-}^{\#1\,\dagger\alpha\beta\chi}$							0	$\frac{1}{2}(Y_{14} + Y_{16} k^2)$

$\mathcal{J}$	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^-}^{\#1}{}_{\alpha\beta\chi}$	$\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{J}_{1^-}^{\#1}{}_{\alpha}$	$\mathcal{J}_{1^-}^{\#2}{}_{\alpha}$	$\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{J}_{2^-}^{\#1}{}_{\alpha\beta\chi}$
$\mathcal{J}_{0^+}^{\#1\,\dagger}$	$\frac{1}{\mathrm{Det}(0^+)}$	0						
$\mathcal{J}_{0^-}^{\#1\,\dagger\alpha\beta\chi}$	0	$\frac{1}{\mathrm{Det}(0^-)}$						
$\mathcal{J}_{1^+}^{\#1\,\dagger\alpha\beta}$			$\frac{Y_8 k^2 + \overset{(2)}{\kappa}_1}{\mathrm{Det}(1^+)}$	$\frac{Y_7 k^2 + 2\,\overset{(2)}{\kappa}_2}{2\sqrt{2}\,\mathrm{Det}(1^+)}$	0	0		
$\mathcal{J}_{1^+}^{\#2\,\dagger\alpha\beta}$			$\frac{Y_7 k^2 + 2\,\overset{(2)}{\kappa}_2}{2\sqrt{2}\,\mathrm{Det}(1^+)}$	$\frac{Y_5 + Y_6 k^2}{2\,\mathrm{Det}(1^+)}$	0	0		
$\mathcal{J}_{1^-}^{\#1\,\dagger\alpha}$			0	0	$\frac{Y_{12} + Y_{13} k^2}{2\,\mathrm{Det}(1^-)}$	$\frac{-Y_{11} k^2 - 2\,\overset{(2)}{\kappa}_3}{2\sqrt{2}\,\mathrm{Det}(1^-)}$		
$\mathcal{J}_{1^-}^{\#2\,\dagger\alpha}$			0	0	$\frac{-Y_{11} k^2 - 2\,\overset{(2)}{\kappa}_3}{2\sqrt{2}\,\mathrm{Det}(1^-)}$	$\frac{Y_9 + Y_{10} k^2}{2\,\mathrm{Det}(1^-)}$		
$\mathcal{J}_{2^+}^{\#1\,\dagger\alpha\beta}$							$\frac{1}{\mathrm{Det}(2^+)}$	0
$\mathcal{J}_{2^-}^{\#1\,\dagger\alpha\beta\chi}$							0	$\frac{1}{\mathrm{Det}(2^-)}$

Abbreviations used in matrices

$$Y_1 == 2\,\overset{(2)}{\kappa}_1 + \overset{(2)}{\kappa}_2 - 3\,\overset{(2)}{\kappa}_3 \;\&\&\; Y_2 == 3\,\overset{(4)}{\kappa}_1 - 2\,\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} + 3\,\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6 \;\&\&$$

$$Y_3 == \overset{(2)}{\kappa}_1 - \overset{(2)}{\kappa}_2 \;\&\&\; Y_4 == \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} \;\&\&\; Y_5 == 2\,\overset{(2)}{\kappa}_1 - \overset{(2)}{\kappa}_2 \;\&\&\; Y_6 == 2\,\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6 \;\&\&$$

$$Y_7 == 2\,\overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_4 \;\&\&\; Y_8 == \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_9 \;\&\&\; Y_9 == 2\,\overset{(2)}{\kappa}_1 + \overset{(2)}{\kappa}_2 - 2\,\overset{(2)}{\kappa}_3 \;\&\&\; Y_{10} == 2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} - 2\,\overset{(4)}{\kappa}_2 \;\&\&$$

$$Y_{11} == 2\,\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_7 \;\&\&\; Y_{12} == 2\,\overset{(2)}{\kappa}_1 + \overset{(2)}{\kappa}_2 - \overset{(2)}{\kappa}_3 \;\&\&\; Y_{13} == 2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_2 - \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_7 + 2\,\overset{(4)}{\kappa}_9 \;\&\&$$

$$Y_{14} == 2\,\overset{(2)}{\kappa}_1 + \overset{(2)}{\kappa}_2 \;\&\&\; Y_{15} == 2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6 \;\&\&\; Y_{16} == 2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} \;\&\&$$

$$\mathrm{Det}(0^+) == \frac{1}{2}\left(-k^2\left(3\,\overset{(4)}{\kappa}_1 - 2\,\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} + 3\,\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6\right) + 2\,\overset{(2)}{\kappa}_1 + \overset{(2)}{\kappa}_2 - 3\,\overset{(2)}{\kappa}_3\right) \;\&\&$$

$$\mathrm{Det}(1^+) == \frac{1}{8}\left(k^4\left(8\,\overset{(4)}{\kappa}_{15}{}^2 - \left(-2\,\overset{(4)}{\kappa}_{16} + \overset{(4)}{\kappa}_4\right)^2 - 4\,\overset{(4)}{\kappa}_{15}\left(\overset{(4)}{\kappa}_{16} + \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6 - 2\,\overset{(4)}{\kappa}_9\right) - 4\left(\overset{(4)}{\kappa}_{16} + \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6\right)\overset{(4)}{\kappa}_9\right) + \right.$$

$$\left. 4\left(\overset{(2)}{\kappa}_1 - \overset{(2)}{\kappa}_2\right)\left(2\,\overset{(2)}{\kappa}_1 + \overset{(2)}{\kappa}_2\right) + 4k^2\left(\left(4\,\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6 + 2\,\overset{(4)}{\kappa}_9\right)\overset{(2)}{\kappa}_1 - \left(\overset{(4)}{\kappa}_{15} + 2\,\overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_9\right)\overset{(2)}{\kappa}_2\right)\right) \;\&\&$$

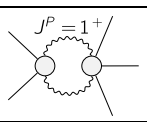
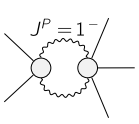
$$\mathrm{Det}(2^+) == \frac{1}{2}\left(k^2\left(2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6\right) + 2\,\overset{(2)}{\kappa}_1 + \overset{(2)}{\kappa}_2\right)$$

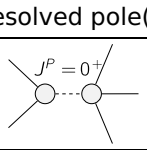
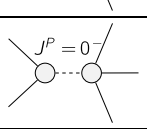
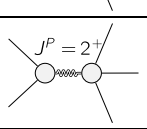
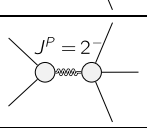
Lagrangian

$$\overset{(2)}{\kappa}_1\,\mathcal{K}_{\alpha\beta\chi}\,\mathcal{K}^{\alpha\beta\chi} + \overset{(2)}{\kappa}_2\,\mathcal{K}^{\alpha\beta\chi}\,\mathcal{K}_{\beta\alpha\chi} + \overset{(2)}{\kappa}_3\,\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\mathcal{K}^{\chi}{}_{\beta\chi} + \overset{(4)}{\kappa}_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} +$$

$$\overset{(4)}{\kappa}_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} + \overset{(4)}{\kappa}_4\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} + \overset{(4)}{\kappa}_6\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} +$$

$$\overset{(4)}{\kappa}_7\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} + \overset{(4)}{\kappa}_9\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi} + \overset{(4)}{\kappa}_{16}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\,\mathcal{J}_{\alpha\beta\chi}$	
Unresolved pole(s)	# d.o.f.	Pole structure(s)
	6	$k^4\left(8\,\overset{(4)}{\kappa}_{15}{}^2 - \left(-2\,\overset{(4)}{\kappa}_{16} + \overset{(4)}{\kappa}_4\right)^2 - 4\,\overset{(4)}{\kappa}_{15}\left(\overset{(4)}{\kappa}_{16} + \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6 - 2\,\overset{(4)}{\kappa}_9\right) - 4\left(\overset{(4)}{\kappa}_{16} + \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6\right)\overset{(4)}{\kappa}_9\right) +$ $4\left(\overset{(2)}{\kappa}_1 - \overset{(2)}{\kappa}_2\right)\left(2\,\overset{(2)}{\kappa}_1 + \overset{(2)}{\kappa}_2\right) + 4k^2\left(\left(4\,\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6 + 2\,\overset{(4)}{\kappa}_9\right)\overset{(2)}{\kappa}_1 - \left(\overset{(4)}{\kappa}_{15} + 2\,\overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_9\right)\overset{(2)}{\kappa}_2\right)$
	6	$k^4\left(8\,\overset{(4)}{\kappa}_{15}{}^2 + 2\,\overset{(4)}{\kappa}_{16}{}^2 - \overset{(4)}{\kappa}_7{}^2 + 4\,\overset{(4)}{\kappa}_2\left(\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 - 2\,\overset{(4)}{\kappa}_9\right) -\right.$ $\left. 2\,\overset{(4)}{\kappa}_{16}\left(3\,\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_7 - 2\,\overset{(4)}{\kappa}_9\right) + 4\,\overset{(4)}{\kappa}_{15}\left(2\,\overset{(4)}{\kappa}_{16} - 3\,\overset{(4)}{\kappa}_2 - \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_7 + 2\,\overset{(4)}{\kappa}_9\right)\right) +$ $2\left(2\,\overset{(2)}{\kappa}_1 + \overset{(2)}{\kappa}_2\right)\left(2\,\overset{(2)}{\kappa}_1 + \overset{(2)}{\kappa}_2 - 3\,\overset{(2)}{\kappa}_3\right) + 2k^2\left(-\left(\left(-4\,\overset{(4)}{\kappa}_{15} - 2\,\overset{(4)}{\kappa}_{16} + 3\,\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_7 - 2\,\overset{(4)}{\kappa}_9\right)\left(2\,\overset{(2)}{\kappa}_1 + \overset{(2)}{\kappa}_2\right)\right) +\right.$ $\left.\left(-6\,\overset{(4)}{\kappa}_{15} - 3\,\overset{(4)}{\kappa}_{16} + 2\left(\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 - 2\,\overset{(4)}{\kappa}_9\right)\right)\overset{(2)}{\kappa}_3\right)$

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$\frac{2\,\overset{(2)}{\kappa}_1 + \overset{(2)}{\kappa}_2 - 3\,\overset{(2)}{\kappa}_3}{3\,\overset{(4)}{\kappa}_1 - 2\,\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} + 3\,\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6}$	$-\frac{2}{3\,\overset{(4)}{\kappa}_1 - 2\,\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} + 3\,\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6}$
	1	$\frac{\overset{(2)}{\kappa}_1 + \overset{(2)}{\kappa}_2}{\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16}}$	$\frac{1}{-\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}}$
	5	$\frac{2\,\overset{(2)}{\kappa}_1 + \overset{(2)}{\kappa}_2}{-2\,\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6}$	$\frac{2}{2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6}$
	5	$-\frac{2\,\overset{(2)}{\kappa}_1 + \overset{(2)}{\kappa}_2}{2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}}$	$-\frac{2}{2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}}$
Resolved unitarity condition(s):	(Demonstrably impossible)		